

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Experienced Embedded Software Engineer with 8+ years developing real-time firmware for ARM Cortex-M microcontrollers in low-power, battery-operated IoT devices. Proven track record in building robust wireless solutions using LoRaWAN, Bluetooth, and NFC. Skilled in power optimization, embedded C, RF integration, and cloud connectivity via AWS and GCP. Strong advocate of test-driven development and CI/CD pipelines, with hands-on experience in the full product lifecycle from prototyping to mass production. Known for cross-functional collaboration and delivering scalable, secure, and energy-efficient embedded systems.

TECHNICAL SKILLS

- **Embedded C**, ARM Cortex-M (especially M-33), FreeRTOS, RTOS
- **IoT Protocols**: LoRaWAN, Bluetooth, NFC, MQTT
- **Wireless & RF**: BLE, Sub-GHz, LPWAN, RF debugging
- **Microcontrollers**: STM32, Nordic nRF52, TI CC13x2
- **Power Optimization**: Battery-operated device firmware, low-power modes, wake-up interrupts
- **Networking**: IoT Architecture, mesh networks, sensor communication
- **Development Tools**: Git, JTAG, ST-Link, GCC, CMake
- **Testing**: Test-Driven Development (TDD), Unit Testing, Hardware-in-the-loop (HIL)
- **Databases**: SQL, NoSQL (MongoDB, Firebase)
- **Cloud Platforms**: AWS (IoT Core, Lambda), GCP, Azure IoT
- **Languages**: C, C++, Python, Bash, TypeScript
- **Other**: PCB Design (basic), CI/CD, Jenkins, Docker

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Architected and programmed firmware in Embedded C for STM32 and Nordic Cortex-M series MCUs powering low-energy IoT sensors with LoRaWAN and BLE communication.
- Designed ultra-low power strategies for battery-operated monitoring devices, achieving up to 24 months of continuous field operation per charge.

- Integrated wireless stacks including Bluetooth 5.0 and LoRaWAN Class A/C with robust retry and acknowledgment mechanisms.
- Developed end-to-end telemetry systems with AWS IoT Core, allowing seamless remote control and real-time monitoring via MQTT.
- Spearheaded RF stack calibration and interference testing across EU/ISM bands using spectrum analysis and TDD-driven iterations.
- Implemented secure firmware-over-the-air (FOTA) updates with rollback and version tracking using NFC boot triggers.
- Collaborated closely with hardware engineers to validate PCB interfaces, ensuring clean ADC/DAC signal integrity and minimal EMI.
- Created CI pipelines for embedded builds using GitLab CI and Dockerized ARM toolchains, reducing release cycles by 35%.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed embedded firmware in C for IoT edge nodes leveraging ARM Cortex-M3 and M4 MCUs, interfaced with industrial sensors.
- Implemented NFC-based device pairing and BLE mesh protocol to streamline deployment in multi-node environments.
- Designed watchdog recovery and fault isolation modules to support continuous uptime in harsh industrial conditions.
- Utilized Git for source management with extensive branching and tagging for production firmware builds.
- Configured UART, SPI, and I2C protocols for communication with environmental sensors and gateway modules.
- Wrote and executed unit and integration tests on physical testbeds, achieving 90% firmware test coverage.
- Collaborated in Agile sprints, delivering critical hotfixes and firmware enhancements during global product rollouts.
- Documented hardware abstraction layers and ISR routines, ensuring seamless developer onboarding and maintenance.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Engineered real-time sensor firmware using C for wearable health IoT devices with ARM Cortex-M0+ MCU platforms.
- Designed ultra-compact sleep-wake loop models with interrupt-based timers for medical-grade energy efficiency.
- Developed BLE service layers for custom device pairing, secure data sync, and GATT profile implementation.
- Integrated cloud connectivity via AWS Lambda and API Gateway to deliver analytics and remote health notifications.
- Used oscilloscope and logic analyzers to trace low-level bus issues across SPI and UART interfaces.
- Led initiative to migrate legacy firmware to modern FreeRTOS-based architecture to support multitasking needs.
- Enabled LoRa-based fallback communication in network-restricted areas to ensure emergency signal coverage.

- Authored manufacturing test firmware for functional QA in production, reducing unit failure rate by 20%.

Software Developer

Decima Digital

November 2015 – April 2016

Tallinn, Estonia

- Participated in firmware design for custom smart home IoT devices with Nordic nRF51822 chips using BLE.
- Configured wake-up interrupts and deep-sleep modes to support 18-month battery life on CR2032 cells.
- Developed companion app interface code in JavaScript for real-time device feedback using Bluetooth GATT.
- Worked with hardware engineers to implement test fixtures for production line functional validation.
- Enhanced EEPROM management and persistent settings layers to allow device personalization and logging.
- Performed packet-level analysis with BLE sniffer tools to debug data corruption under low-signal environments.
- Implemented cyclic redundancy checks (CRC) and retry queues for robust sensor-to-gateway transmission.
- Built automated test cases using Python scripts for smoke testing new firmware versions on devices.

Software Developer Intern

Borna Technology

September 2014 – August 2015

Tallinn, Estonia (On-Site)

- Assisted in firmware development for sensor-based monitoring systems using PIC microcontrollers and Embedded C.
- Contributed to driver development for analog sensors including temperature and current sensing modules.
- Designed serial command interfaces using UART for device configuration and data streaming.
- Integrated simple real-time task schedulers to handle event-based processing with timing precision.
- Created simulation models to validate ADC resolution against hardware filtering parameters.
- Ported legacy 8-bit microcontroller code to newer 32-bit architecture with minimal code footprint increase.
- Documented software testing procedures and assisted QA team with functional validation.
- Participated in weekly cross-functional meetings and provided firmware debugging assistance on-site.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ✓ **Battery-Optimized LoRaWAN Asset Tracker (2023)**
Built ultra-low power asset tracker using STM32L4 series and LoRa radio. Implemented deep sleep scheduling, motion-triggered wakeup, and LoRaWAN Class A with ADR. Achieved >18 months operation per battery.
 - ✓ **BLE + NFC Smart Sensor Network (2022)**
Designed firmware for smart HVAC sensors with BLE for mesh data transmission and NFC for quick configuration. Integrated device provisioning with NFC tags and BLE-to-Wi-Fi gateways.
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PROJECTS

- Embedded Systems Essentials with Arm (Coursera, 2023)
- LoRaWAN for Developers (The Things Network, 2023)
- Arm Cortex-M Microcontroller Development (Udemy, 2022)
- AWS IoT Core for Embedded Systems (LinkedIn Learning, 2024)